

FLYING IN ANCIENT EGYPT



Fig. 1- The side view of the small model explaining the fuselage and the vertical tail.



Fig. 2- The front view of the small model explaining the dihedral angle of the wings.

A small wooden artifact was found in the excavations around the step pyramid of Saqqara in 1898 by the French archaeologist Lauret. This piece was registered in the archives of the Egyptian museum of Antiquities in Cairo as a "Statuette of a bird". It was placed in the bird section of the museum under the number 6347, and a little more was thought of it for 70 years.

The Egyptian Physician, artist and aeromodeller [Dr. Khalil Messiha](#) rediscovered the same artifact in the year 1969 in the museum. [Dr. Khalil](#) found that this model differs considerably from other birds models in that it was legless and had a straight tail and wings. The figure weighed 39.12 gms, its wings were 18 cm long and its fuselage is a 14 -cm long piece with an airfoil shape beautifully carved and smooth except for the tip which is slightly worn out. The part of the tail is broken, and Dr. Khalil presumes that a stabilizer was attached to that missing part. The aging wood artifact was affected by weather conditions. The vertical tail is slightly slanting towards the right when looked at from the back.

Dr. Khalil made a balsa wood model with the same measurements as the Pa-Di-Imen artifact and added a stabilizer to the tail. the model was pushed by hand and flew a few yards. Some call it a bird,... but no birds with vertical tails were ever seen in Egypt on the middle East. The museum artifact has no legs like the thousands of birds statuettes found in the excavations in Egypt. Some think it is a toy, but Ancient Egyptian technologists always made models of things they were going to build, all the way from ships to temples. Egypt's history of flight is thousands of years old, and first manifested itself in the philology of the Ancient Egyptian language and the religious rites practiced by the Ancient Egyptians. In Ancient Egyptian language there were three words which meant to fly "hi-, hi, and pa", the determinative in all three being a half - wing with long feathers. In the Egyptian museum of Cairo there are a variety of birds scenes depicted on murals since the Old Kingdom. Moreover, the God Horus - as a falcon and Horus - Ra the son of Osiris as a falcon in the sun disc on his head - were favorite themes both in portraiture and sculpture. "Outstretched wings" are a feature of a great number of statuettes of the Cairo Egyptian Museum, among which are winged scarabs from the Greco Roman period found in Upper Egypt, and a fine bronze figure of Nefertum, son of Ptah and Sekhmet dating from (600 BC.). We hope that one day we can find Ancient Egyptian aeroplane capable to carry young man and floats in the sky.



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